

# ModelNet40 Point-Cloud Upsampling and PointNet++ Classification: Complete Experimental Dossier

Purpose: Organize the research record and prepare the thesis and defence.

Original project root: /home/woody/iwnt/iwnt189h/thesis\_pointcloud/modelnet40\_pointnet2\_upsampling

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Primary numerical sources: pointnet2\_results/x4\_two\_line\_final/\*\*/metrics.json,  
reports/modelnet40\_geometry\_equal\_n\_\*,  
reports/modelnet40\_pointnet2\_final\_two\_line\_classification\_summary\_with\_pu\_edgeformer.\*

Companion PDF: ModelNet40\_Thesis\_Complete\_Dossier.pdf. Paths below refer to the original project unless stated otherwise; see the archive README for the current directory layout.

## Main finding

Under a common  $\times 4$  two-line protocol on ModelNet40, five upsampling methods were compared with PointNet++ trained from scratch for each input variant. Better geometry did not automatically improve classification: every Line A (densification) result was below Original 1024, and only PU-Net in Line B (sparse-input recovery) slightly exceeded the downsampled baseline (+0.42 pp), while remaining below Original.

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## 1. Background and research questions

Point-cloud upsampling aims to produce a denser, geometrically more complete point set from a sparse point cloud. Much of the literature evaluates Chamfer Distance (CD), Hausdorff Distance (HD), and point-to-surface distance (P2F), assuming that increased density and better geometry benefit downstream three-dimensional tasks.

This project tests that assumption under the official ModelNet40 classification setting: does Overall Accuracy (OA) improve when the upsampled output is supplied to PointNet++?

## 1.1 Research questions

RQ1: Does  $\times 4$  densification of an already well-sampled Original 1024 input to 4096 points improve PointNet++ classification?

RQ2: After  $\times 4$  downsampling to 256 points, can recovery to 1024 points recover lost classification performance?

RQ3: Is the method with the best geometric metrics also the best for downstream classification?

RQ4: Can increased point count alone, tested by directly sampling 4096 mesh points, explain the classification changes?

## 1.2 Scope

| Item             | Setting                                                                                                    |
|------------------|------------------------------------------------------------------------------------------------------------|
| Task             | ModelNet40 shape classification; not KITTI detection or AP                                                 |
| Downstream model | PointNet++ SSG (pointnet2_cls_ssg), XYZ input only                                                         |
| Upsamplers       | Inference with existing pretrained weights; the upsampling networks are not retrained in these experiments |
| Classifier       | Trained from scratch for each point-count setting, without fine-tuning the 1024-point weights              |
| Main ratio       | $R=\times 4$                                                                                               |
| TULIP            | Range-image / LiDAR method; supplementary only, excluded from the main xyz comparison                      |

Contribution: Comparing five methods under the same  $\times 4$  two-line protocol shows why geometric recovery and classification utility must be reported separately. Densification did not help in this setting; only PU-Net produced a marginal recovery benefit, still below Original.

## 2. Evaluation principles

### 2.1 Two-line design

The two experimental lines distinguish densification of a well-sampled input from recovery of a sparse input. Cross-line comparisons do not establish the main conclusions.

| Experimental line    | Purpose                      | Input path                                                               | PointNet++ input points |
|----------------------|------------------------------|--------------------------------------------------------------------------|-------------------------|
| Line A Densification | Densify well-sampled data    | Original 1024 $\rightarrow$ Up $\times 4$                                | baseline 1024 / up 4096 |
| Line B Recovery      | Recover after sparsification | Original $\rightarrow$ Down $\times 4$ (256) $\rightarrow$ Up $\times 4$ | baseline 256 / up 1024  |

### 2.2 Mandatory point-count protocol

1. Keep each baseline at its native point count: do not pad or resample it to the upsampled output size.
2. Keep each upsampled output at its output point count: do not crop it back to the baseline size.
3. Do not artificially match baseline and upsampled point counts; doing so confounds the upsampling effect with the point-count effect.
4. Within a line and branch, use the same upsampling ratio and output point count.
5. The Original and Downsampled lines are independent; there is no requirement to match point counts across lines.

### 2.3 Classification metrics

Dataset: the official ModelNet40 split, 9843 train / 2468 test, with no independent validation set for classification, following the recorded PointNet++ setup.

Best Overall Accuracy is the highest overall (instance) accuracy observed during training. This implementation saves the best checkpoint when test accuracy improves.

Final Overall is the overall accuracy at epoch 200, reported as supplementary information.

Best OA is the primary reported metric; differences are in percentage points (pp).

For Line B, the primary  $\Delta$  is relative to the Downsampled  $\times 4$  baseline; the secondary gap is relative to Original.

## 2.4 Geometric metrics: distinct reference protocols

| Version                           | Reference                                                   | Role                       | Interpretation                                                            |
|-----------------------------------|-------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------|
| Equal-N (current)                 | A: mesh-ref 4096; B down: mesh-ref 256; B up: Original 1024 | Main geometry table        | Use equal-cardinality CD/HD for the intended comparison                   |
| vs Original (unequal cardinality) | 4096 vs 1024, etc.                                          | Historical / supplementary | Measures agreement between discrete sets, not continuous-surface accuracy |
| Earlier mesh / exact P2F          | dense mesh surface                                          | Computed in earlier work   | Removed from the main narrative in the 2026-08 PPT revision               |

Metric definitions:

CD (Chamfer): the sum of mean nearest-neighbor distances in both directions, measuring overall agreement.

HD (Hausdorff): the worst nearest-neighbor discrepancy, sensitive to outliers.

NUC: point-distribution non-uniformity; use relative comparisons within the protocol, with lower values generally indicating greater uniformity.

Mesh-ref sampling: triangle area-weighted sampling from .off meshes followed by unit-sphere normalization; `seed=stable_seed(42, ...)`.

Mesh-ref is sampled independently from the same mesh family, rather than downsampled from Original 1024.

## 3. Canonical experimental protocol

The main protocol was finalized from 2026-07 and supplemented with equal-N / mesh-ref baselines in 2026-08. Historical  $\times 2$  / 512-point experiments are ablations only; see Section 11.

| Line | Branch                                                | Points on disk | num_point | allow_resample | Role                        |
|------|-------------------------------------------------------|----------------|-----------|----------------|-----------------------------|
| A    | Original baseline                                     | 1024           | 1024      | false          | Main baseline               |
| A    | Original + EAR / PDANS / PU-Net / PU-GCN / EdgeFormer | 4096           | 4096      | false          | Main upsampling             |
| B    | Downsampled $\times 4$ baseline                       | 256            | 256       | false          | Main baseline               |
| B    | Down $\times 4$ + the same five methods               | 1024           | 1024      | false          | Main upsampling             |
| —    | Mesh-ref 256                                          | 256            | 256       | false          | Sampling-process control    |
| —    | Mesh-ref 4096                                         | 4096           | 4096      | false          | Dense mesh-sampling control |

## Shared training settings

| Item                    | Value                                               |
|-------------------------|-----------------------------------------------------|
| Model                   | pointnet2_cls_ssg                                   |
| Number of classes       | 40                                                  |
| Input features          | XYZ only                                            |
| epoch                   | 200                                                 |
| optimizer               | Adam                                                |
| learning_rate           | 0.001                                               |
| decay_rate              | 0.0001                                              |
| batch_size              | 24                                                  |
| seed                    | 42                                                  |
| num_workers             | 4                                                   |
| Configuration directory | configs/pointnet2_x4_two_line/*.yaml                |
| Results directory       | pointnet2_results/x4_two_line_final/**/metrics.json |

## 4. Methods and technical background

| Method        | Type                             | Principle                                                                                                                                             | Paper / source                                                                        |
|---------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| EAR           | Classical geometry, non-learning | Samples away from sharp features to estimate reliable normals, then progressively upsamples near edges; preserves sharp features and controls density | Huang et al., ACM TOG 2013 — *Edge-aware point set resampling*                        |
| PU-Net        | Learned (TF1)                    | Multi-level point features and multi-branch feature-space expansion; a pioneering learned point-cloud upsampler                                       | Yu et al., CVPR 2018 — arXiv:1801.06761                                               |
| PU-GCN        | Learned (TF1)                    | Graph convolutions model local geometric relationships; often performs well on CD                                                                     | Qian et al., CVPR 2021 — arXiv:1912.03264                                             |
| PDANS         | Learned (PyTorch)                | Conditional diffusion upsampling with Adaptive Noise Suppression / Tree-Trans, emphasizing noise robustness and detail                                | CVPR 2025 — <a href="https://github.com/Baty2023/PDANS">github.com/Baty2023/PDANS</a> |
| PU-EdgeFormer | Learned (TF)                     | EdgeFormer combines local EdgeConv with global multi-head self-attention                                                                              | Kim et al., ICASSP/arXiv 2023 — arXiv:2305.01148                                      |
| PointNet++    | Downstream classification        | Hierarchical sampling and local PointNet aggregation; SSG is used here                                                                                | Qi et al., NeurIPS 2017; yanx27 PyTorch implementation                                |
| ModelNet40    | Dataset                          | Point clouds sampled from CAD meshes, with the official train/test split                                                                              | Wu et al., CVPR 2015                                                                  |
| TULIP         | Supplementary only               | LiDAR range-image upsampling; incompatible with the main xyz point-count comparison                                                                   | ETHZ ASL TULIP; excluded from the main table                                          |

### Implementation and inference notes

EAR: deterministic script without a checkpoint; migrated from the lab to HPC.

PU-Net / PU-GCN / PU-EdgeFormer: TensorFlow with custom CUDA/tf\_ops; matching build environments are required.

PDANS: PyTorch with PyTorch3D / Chamfer / pointops and large .pkl checkpoints (PU1K / PUGAN).

PU-EdgeFormer: integrated on 2026-07-16, with an authenticity audit covering the model path, checkpoint, differences from PU-GCN output, and absence of leakage.

All main methods generated the complete 12311 samples (9843+2468) on both lines before point-count / NaN checks and PointNet++ evaluation.

## 5. Classification results

### 5.1 Line A — Densification relative to Original 1024

| Method            | Points | Best OA | Final OA | Best Class | $\Delta$ Best vs Orig (pp) |
|-------------------|--------|---------|----------|------------|----------------------------|
| Original baseline | 1024   | 91.95%  | 91.31%   | 88.00%     | 0.00                       |
| PU-GCN            | 4096   | 91.63%  | 91.00%   | 87.74%     | -0.32                      |
| PDANS             | 4096   | 91.62%  | 90.87%   | 88.44%     | -0.33                      |
| EAR               | 4096   | 91.48%  | 90.85%   | 88.19%     | -0.47                      |
| PU-Net            | 4096   | 90.95%  | 90.55%   | 87.46%     | -1.00                      |
| PU-EdgeFormer     | 4096   | 90.54%  | 90.13%   | 86.35%     | -1.41                      |

Interpretation: No 4096-point upsampled branch exceeded Original 1024. PU-GCN (-0.32 pp) and PDANS (-0.33 pp) were closest, while PU-EdgeFormer had the largest decline (-1.41 pp). In this setting, densifying already well-sampled CAD point clouds did not reliably improve PointNet++ classification.

### 5.2 Line B — Recovery relative to Down 256, with the gap to Original

| Method                 | Points | Best OA | Final OA | $\Delta$ vs Down (pp) | gap vs Orig (pp) |
|------------------------|--------|---------|----------|-----------------------|------------------|
| Downsampled $\times 4$ | 256    | 90.85%  | 90.46%   | 0.00                  | -1.10            |
| PU-Net                 | 1024   | 91.27%  | 90.65%   | +0.42                 | -0.68            |
| PDANS                  | 1024   | 90.34%  | 89.46%   | -0.51                 | -1.61            |
| PU-GCN                 | 1024   | 90.06%  | 89.63%   | -0.79                 | -1.89            |
| PU-EdgeFormer          | 1024   | 89.32%  | 88.58%   | -1.53                 | -2.63            |
| EAR                    | 1024   | 88.81%  | 88.02%   | -2.04                 | -3.14            |

Interpretation: Restoring point count did not automatically restore classification performance. Only PU-Net improved over the sparse baseline (+0.42 pp), while remaining 0.68 pp below Original. EAR had the weakest classification result (-2.04 pp). Returning to 1024 points therefore did not necessarily restore semantic separability.

## 6. Equal-N geometry results

Split: test, N=2468

Script: `scripts/compute_geometry_equal_n.py`

Outputs: `reports/modelnet40_geometry_equal_n_*. {csv,md,json}`

### 6.1 Line A: 4096 vs Mesh-ref 4096; lower is better

| Method               | CD       | HD       | NUC      | $\Delta$ NUC |
|----------------------|----------|----------|----------|--------------|
| Mesh-ref 4096 (self) | 0.000000 | 0.000000 | 0.641852 | 0.000        |
| PU-GCN               | 0.042306 | 0.093219 | 0.520398 | -0.121       |
| PU-EdgeFormer        | 0.046666 | 0.110932 | 0.675435 | +0.034       |
| PDANS                | 0.046821 | 0.113510 | 0.661176 | +0.019       |
| PU-Net               | 0.049719 | 0.111177 | 0.605211 | -0.037       |
| EAR                  | 0.054722 | 0.115396 | 0.815631 | +0.174       |

## 6.2 Line B: Down 256 vs Mesh-ref 256; Ups 1024 vs Original 1024

| Method              | Points | Reference     | CD     | HD     | NUC   |
|---------------------|--------|---------------|--------|--------|-------|
| Mesh-ref 256 (self) | 256    | Mesh-ref 256  | 0.000  | 0.000  | 2.126 |
| Downsampled ×4      | 256    | Mesh-ref 256  | 0.133  | 0.205  | 2.093 |
| Original (self)     | 1024   | Original 1024 | 0.000  | 0.000  | 1.084 |
| PU-GCN              | 1024   | Original 1024 | 0.0558 | 0.1514 | 1.524 |
| PU-EdgeFormer       | 1024   | Original 1024 | 0.0619 | 0.1776 | 1.123 |
| PDANS               | 1024   | Original 1024 | 0.0627 | 0.1905 | 1.053 |
| PU-Net              | 1024   | Original 1024 | 0.0646 | 0.1083 | 1.080 |
| EAR                 | 1024   | Original 1024 | 0.0767 | 0.1932 | 0.965 |

Key observations: PU-GCN generally had the best CD on both lines, but its Line B NUC deteriorated to 1.524. PU-Net had the best Line B HD (0.108) and NUC close to Original, consistent with its unique classification gain. EAR could have a lower NUC while still performing poorly on CD/HD and classification.

Note: Unequal-cardinality CD/HD comparisons (4096 vs 1024 and 256 vs 1024) are deliberately excluded from the main ranking.

## 7. Geometry and classification

| Observation                                          | Evidence                                                           | Implication                                                 |
|------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------|
| Better geometry does not imply better classification | In Line B, PU-GCN has the lowest CD, but lower Best OA than PU-Net | Report geometry and task metrics separately                 |
| Densification did not help                           | All Line A 4096-point methods are below Original 1024              | The densification hypothesis is unsupported in this setting |
| Recovery is limited                                  | Only PU-Net gains +0.42 pp and remains below Original              | Upsampling does not fully offset the ×4 information loss    |
| Distribution matters                                 | PU-Net has more balanced HD/NUC; PU-GCN NUC deteriorates           | PointNet++ is sensitive to local structure and distribution |
| Point count alone is a weak explanation              | Mesh-ref 4096 is only +0.05 pp above Original                      | Increased density alone does not explain the effect         |

## 8. Mesh-ref controls

To examine whether classification changes arise merely from point count or sampling procedure, mesh-ref 256 and mesh-ref 4096 were independently sampled from .off meshes with area weighting. PointNet++ was trained from scratch with the same hyperparameters.

| Method        | Points | Best OA | Final OA | best epoch | Control       | Control Best | Δ pp  |
|---------------|--------|---------|----------|------------|---------------|--------------|-------|
| Mesh-ref 256  | 256    | 90.88%  | 90.38%   | 129        | Down ×4 256   | 90.85%       | +0.03 |
| Mesh-ref 4096 | 4096   | 92.00%  | 91.18%   | 148        | Original 1024 | 91.95%       | +0.05 |

Interpretation:

1. At the same point count, direct mesh sampling and downsampling Original gave nearly identical classification (+0.03 pp).
2. Denser mesh-ref 4096 was only +0.05 pp above Original 1024, with no substantial advantage.
3. The consistent deficit of Line A upsampled branches is therefore more plausibly associated with their geometric / distributional properties than with the number 4096 itself.

## 9. Protocol history

| Stage                   | Change                                                                                                              | Current status                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Early work              | KITTI + CenterPoint/PointRCNN detection; migration of methods and data from lab to HPC                              | Parallel engineering background, outside the main results here |
| 2026-06-15              | Established modelnet40_experiments and method_registry                                                              | Asset index retained                                           |
| 2026-06-20+             | Built the main modelnet40_pointnet2_upsampling pipeline                                                             | Current project root                                           |
| 2026-06-25              | Main ratio changed from $\times 2$ to $\times 4$ ; earlier 512 $\rightarrow$ 1024 EAR experiment became an ablation | Implemented                                                    |
| Subsequent finalization | Line B changed to 256 $\rightarrow$ 1024 recovery; 512 $\rightarrow$ 2048 ceased to be the main line                | Canonical                                                      |
| Implementation          | Fixed CUDA/TF ops, the PDANS environment, PU-Net timeout/resume, and PU-GCN stalls                                  | See reports/                                                   |
| 2026-07-16              | Integrated PU-EdgeFormer after a successful authenticity audit                                                      | Fifth method                                                   |
| 2026-07-12~17           | Finalized thesis tables, figures and interpretations; clarified geometry vs classification                          | Basis of the results chapter                                   |
| 2026-08-03              | Recomputed equal-N mesh-ref geometry and completed mesh-ref classifier training                                     | Main geometry protocol                                         |
| 2026-08-04              | Removed the main mesh/P2F narrative from the PPT; prepared Full Report and emphasized training from scratch         | Presentation version                                           |

### Corrections relevant to limitations and appendices

1.  $R=\times 2$  was briefly recommended, then changed to  $\times 4$  to match the default of the PU methods and the experimental objective.
2. Early Line B used downsampled50 (about 512 points), later replaced by downsampled\_x4 (256 points) for strict  $\times 4$  recovery.
3. The baseline loader was previously resampled to 1024 in Step 6. This is now classified as a naive ablation, not a main baseline.
4. The geometric narrative changed from dense mesh + P2F to equal-N, avoiding misleading rankings from unequal-cardinality CD/HD.
5. Best OA is explicitly defined as the highest recorded checkpoint overall accuracy, without a separate validation set.

## 10. Implementation and work record

### 10.1 Data and generation

Original 1024 point clouds were sampled from ModelNet40 meshes and normalized to the unit sphere.

Downsampled  $\times 4$ : datasets/modelnet40\_downsampled\_x4/ (256 points, 12311 shapes).

Line A/B outputs passed strict point-count checks (4096 or 1024 expected) with zero NaN/Inf values.

PointNet++ inputs were organized under pointnet2\_inputs/ using symbolic links and preparation scripts.

### 10.2 Failures and repairs

PDANS: an early job failed with `torch.cuda.is_available()=False`; aligning modules and `LD_LIBRARY_PATH` resolved the issue.

PU-Net / PU-GCN: TF ops depend on compatible CUDA build and runtime environments.

PU-Net Line B full: multiple chunks reached the 24h timeout; resuming the missing 3161 samples eventually produced a 12311/12311 pass.

PU-GCN: an RTX 3080 stall and a SyntaxError were resolved through repairs and resume runs.

PU-EdgeFormer: separate audits covered repeat-copy behavior, call paths, checkpoints, differences from PU-GCN, leakage, and output freshness.

### 10.3 Figures and presentation materials

Static figures: figures/modelnet40/pointnet2\_final\*\_with\_pu\_edgeformer/, geometry\_delta\_\*, final\_combined/.

Interactive HTML: figures/modelnet40/pointcloud\_examples\_interactive\_v2/dropdown\_v2/ (10 dropdown views).

PPT: ModelNet40\_PointNet2\_Full\_Report.pptx (28 slides) and Original\_Reference\_Revised.pptx (21 slides).

Matched elevation, azimuth, axis limits, and point-size rules support fair qualitative comparisons between methods.

## 11. Ablations and supplementary results

| Item                              | Content                                                       | Role                                                           |
|-----------------------------------|---------------------------------------------------------------|----------------------------------------------------------------|
| ×2 EAR Line B                     | 512→1024, earlier Step 7b/8                                   | Ratio ablation, outside the main ×4 protocol                   |
| Naive resampling baseline         | 512 points on disk, resampled by the loader to 1024 in Step 6 | Illustrates contamination from artificial point-count matching |
| Earlier downsampled50 main line   | 512-point protocol                                            | Replaced by ×4/256                                             |
| TULIP                             | Range-image upsampling                                        | Supplementary only                                             |
| Unequal-cardinality CD/HD ranking | 4096 vs 1024, etc.                                            | Excluded from the main geometric ranking                       |
| Earlier P2F narrative             | Dense mesh exact P2F                                          | Appendix material, outside the current main table              |
| KITTI / CenterPoint               | Detection AP experiments                                      | Separate chapter, not expanded in this dossier                 |

## 12. Thesis structure and usable conclusions

### Suggested structure

1. Introduction: motivate systematic downstream classification evaluation beyond geometric metrics and introduce the two-line ×4 question.
2. Related Work: EAR → PU-Net → PU-GCN → EdgeFormer / diffusion (PDANS), PointNet++, and ModelNet40.
3. Method: protocol principles, two-line processing, classification from scratch, and equal-N geometry.
4. Experimental Setup: data, hyperparameters, metric definitions, and implementation details.
5. Results: classification, equal-N geometry, and mesh-ref tables; report absolute OA and  $\Delta$  pp together.
6. Discussion: the gap between geometry and task utility, densification vs recovery, and the roles of distribution, HD, and NUC.
7. Ablation / Appendix: ×2, naive resampling, mesh-ref controls, and engineering audits.
8. Conclusion: the main findings and implications listed below.

### Conclusions for adaptation into the thesis

1. Point-cloud upsampling does not generally improve PointNet++ classification accuracy on ModelNet40 in these experiments.
2. Under densification (1024→4096), all five methods have lower Best OA than native Original 1024 (91.95%).
3. Under recovery (256→1024), only PU-Net improves over the sparse baseline (+0.42 pp), while remaining 0.68 pp below Original.

4. Geometric metrics and classification utility differ: PU-GCN often has better CD but weaker classification than PU-Net.
5. Mesh-ref 4096 is only +0.05 pp above Original, showing that increased point count alone cannot explain or guarantee classification gains.
6. Upsampling evaluations should report geometric quality and downstream utility together, explicitly distinguishing densification from recovery.

## 13. File index

| Purpose                             | Path relative to the original project root                                              |
|-------------------------------------|-----------------------------------------------------------------------------------------|
| This Markdown dossier               | presentations/ModelNet40_Thesis_Complete_Dossier.md                                     |
| Companion PDF                       | presentations/ModelNet40_Thesis_Complete_Dossier.pdf                                    |
| Full PPT/PDF                        | presentations/ModelNet40_PointNet2_Full_Report.*                                        |
| Revised PPT                         | presentations/ModelNet40_PointNet2_Original_Reference_Revised.*                         |
| Classification summary              | reports/modelnet40_pointnet2_final_two_line_classification_summary_with_pu_edgeformer.* |
| Geometry and classification summary | reports/modelnet40_final_thesis_summary_table_with_pu_edgeformer.*                      |
| Interpretation                      | reports/modelnet40_thesis_result_analysis.md                                            |
| Equal-N geometry                    | reports/modelnet40_geometry_equal_n_*. {csv,md,json}                                    |
| Mesh-ref classification             | reports/modelnet40_pointnet2_mesh_ref_baseline_results.*                                |
| x4 protocol report                  | reports/modelnet40_x4_final_protocol_report.md                                          |
| Protocol correction log             | reports/modelnet40_pointnet2_final_protocol_corrected.md                                |
| EdgeFormer audit                    | reports/pu_edgeformer_modelnet40_authenticity_verification_20260716.md                  |
| Figure captions                     | reports/modelnet40_thesis_figure_captions.md                                            |
| Figure index                        | figures/modelnet40/figure_index.md                                                      |
| Method registry                     | ../modelnet40_experiments/configs/method_registry.yaml                                  |
| Training configurations             | configs/pointnet2_x4_two_line/*.yaml                                                    |
| Metrics JSON                        | pointnet2_results/x4_two_line_final/**/metrics.json                                     |

## Appendix A. Training settings and data paths

| Item                      | Value                                    |
|---------------------------|------------------------------------------|
| Data split                | 9843 train / 2468 test, official split   |
| Model                     | pointnet2_cls_ssg                        |
| epoch / batch / lr / seed | 200 / 24 / 0.001 / 42                    |
| optimizer / decay         | Adam / 0.0001                            |
| Features                  | XYZ only                                 |
| Line A data               | pointnet2_inputs/lineA_*                 |
| Line B data               | pointnet2_inputs/lineB_*                 |
| Downsampled root          | datasets/modelnet40_downsampled_x4/      |
| Mesh-ref roots            | datasets/modelnet40_mesh_ref_{256,4096}/ |
| Results root              | pointnet2_results/x4_two_line_final/     |

## Appendix B. Code and checkpoint sources

| Method        | Code source                                    | Weights / notes                                               |
|---------------|------------------------------------------------|---------------------------------------------------------------|
| EAR           | code/upsampling_methods/EAR, migrated from lab | No checkpoint; non-learning method                            |
| PU-Net        | yulequan/PU-Net, migrated                      | TF1 model-120, etc.                                           |
| PU-GCN        | guochengqian/PU-GCN, migrated                  | TF1 model-120, etc.                                           |
| PDANS         | Baty2023/PDANS, migrated                       | PU1K_PDANS.pkl / PUGAN_PDANS.pkl                              |
| PU-EdgeFormer | external/pu_edgeformer_ops_reuse               | checkpoint_model100; audited 2026-07-16                       |
| TULIP         | ethz-asl/TULIP, migrated                       | tulip_kitti.pth; supplementary                                |
| PointNet++    | external/Pointnet_Pointnet2_pytorch            | All classifier training in these experiments was from scratch |

## Document notes

Where earlier Markdown conflicts with this record, the 2026-08 Full Report / equal-N / classification\_summary\_with\_pu\_edgeformer takes precedence for this dossier.

KITTI / CenterPoint detection belongs to earlier or parallel work. A detection chapter requires the separate centerpoint\_variant\_eval and KITTI upsampled\_variants records.

Regenerate the companion PDF from the archive root with python  
research/modelnet40/code/scripts/generate\_thesis\_complete\_dossier\_pdf.py.